

Tatiana Acero-Cuellar

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Ph.D. Physics Candidate at the University of Delaware | Machine Learning, Astrophysics, Geospatial Data Science
| taceroc.github.io | github.com/taceroc | [Google Scholar](https://scholar.google.com/citations?user=8Q8Q8Q8Q8Q) | [linkedin.com/in/tatiana-acero-cuellar](https://www.linkedin.com/in/tatiana-acero-cuellar) |

Summary

I have direct experience on the Vera C. Rubin Observatory Alert Production commissioning team. I developed and implemented the Real-Bogus deep learning model for Rubin, assigning a reliability score to detected sources to distinguish astrophysical transients from artifacts. I led the Rubin Difference Detectives, a citizen science project on the Zooniverse platform. My research in astrophysics includes developing novel methods for transient discovery and physics-driven simulations of light echoes in astrophysical images. Outside of astrophysics, I have led work on the automatic detection of fish ponds in Nigeria using satellite imagery. I am skilled in Python, PyTorch, TensorFlow, computer vision, and interdisciplinary collaboration. All projects I have developed are stored and versioned using GitHub. I have a strong background in analytical thinking and problem-solving. I bring both independent research experience and the ability to work effectively in small and large teams. I am eager to learn new skills, self-motivated, and driven to research and develop novel solutions to problems. I am responsible, disciplined, and determined, consistently giving my full effort.

Skills

Python, NumPy, Keras, Seaborn, scikit-learn, Pandas, TensorFlow, PyTorch, GeoPandas, Rasterio, OpenCV, UMAP, SciPy, Xarray, ArcGIS, QGIS, GitHub, Hugging Face, Zenodo. Machine learning, Deep Learning, Computer vision, Remote Sensing.

Education

Ph.D. Physics, Department of Physics and Astronomy, University of Delaware, Newark, DE GPA: 3.9/4.0	August 2022 - present
Geospatial Data Science Certificate, Department of Geography and Spatial Science, University of Delaware, Newark, DE	August 2024 – Dec 2025
B.Sc., Physics Universidad Nacional de Colombia, Bogotá, D.C GPA: 4.5/5.0	August 2015 - April 2021

Publications

- **Acero-Cuellar, Tatiana.**, et al. 2026, “The Real-Bogus Machine Learning Model for the Vera C. Rubin Observatory”, in preparation.
- Weston, J.G., **Acero-Cuellar, Tatiana.**, et al. 2026, “Identifying Expert Labelling for Rubin Difference Detectives”, in preparation.
- **Acero-Cuellar, Tatiana.**, Bianco, Federica., Bellm, Eric., Sako, Masao., Sanchez, Bruno. 2026, “Performance of Machine-Learned Reliability Scoring for Image Differencing”, Data Management Technical Note DMTN-337, NSF-DOE Vera C. Rubin Observatory. <https://dmtn-337.lsst.io>
- **Acero-Cuellar, Tatiana.**, Ayinde, Rodiat., Upon, John., et al. 2026. “Automated detection and segmentation of smallholder fish ponds in Nigeria”, PREPRINT (Version 1) available at Research Square. <https://doi.org/10.21203/rs.3.rs-8613311/v1> (accepted for publication in Scientific Data)

- Inada, Adi., Sako, Masao., **Acero-Cuellar, Tatiana.**, and Bianco, Federica. 2025. "Transformer-Based Neural Network for Transient Detection without Image Subtraction." The Astronomical Journal 171 (4): 205. <https://doi.org/10.3847/1538-3881/ae38d8>.
- Vera C. Rubin Observatory Team, **Acero-Cuellar, Tatiana.**, Acosta, Emily, et al. 2025. "RTN-095: The Vera C. Rubin Observatory Data Preview 1", <https://doi.org/10.71929/rubin/2570536>
- **Acero-Cuellar, Tatiana.**, Bianco, Federica., Dobler, Gregory., Sako, Masao., and Qu, Helen. 2023. "What's the Difference? The Potential for Convolutional Neural Networks for Transient Detection without Template Subtraction." The Astronomical Journal 166 (3): 115. <https://doi.org/10.3847/1538-3881/ace9d8>.

Scholarships and Fellowships

- **Unidel Distinguished Grad Scholar Fellowship from the Graduate College, University of Delaware; 2022-2027 (competitive, 5-year funding)**
- LSST Data Science Fellowship Program 2022-2024 (competitive, full scholarship)
- LSST Corporation and the Enabling Science Grant program partially supported TAC through Grant No. 2021-040 (Artificial Intelligence Solutions to Transient Discovery in Rubin LSST)
- University of Delaware Summer Research Program; July 2020 - December 2021 (competitive)

Ph.D. Research

Implementation of the Real-Bogus Deep Learning model for the NSF-DOE Vera C. Rubin Observatory. (2024-present)

- Developed and implemented a Convolutional Neural Network (CNN) that assigns each detected source a reliability score in $[0,1]$, where 1 indicates a real astrophysical source, and 0 indicates an artifact.
- Designed the model to meet Rubin Observatory's strict performance and computational requirements using a supervised training strategy in PyTorch.
- Collected and curated training data by cross-matching Rubin-detected sources with external catalogs (Gaia variables, Transient Name Server, Rubin Solar System objects).
- Worked with the LSST Science Pipeline components to extract data for training and validation.
- Trained and validated the model meeting the performance and computing time requirements established by the Rubin Observatory.
- Documented the performance of all versions of the model in a Data Management Technical note.
- Tested different training strategies (e.g., training with noisy labels, contrastive learning training, informative embedding space) to improve the model's performance.

Citizen Science: Rubin Difference Detectives at Zooniverse platform (2025-present)

- Launched a Zooniverse project where volunteers classified Rubin images as real astrophysical sources or artifacts.
- Selected sources to ensure variety by cross-matching with the Transient Name Server, Gaia variables catalog, and Rubin Solar System objects catalog.
- Designed and wrote the question and tutorial provided to volunteers.
- Analyzed volunteers' responses and transformed them into ready-to-use data for machine learning training.
- Used volunteer labels to retrain the Real-Bogus model, directly improving classification performance.

A national fish pond dataset for Nigeria generated using automated detection and segmentation (2024-2026)

- Developed and implemented a multistage deep-learning+random forest pipeline.
- Implemented and fine-tuned the computer vision pre-trained model YOLOv7.
- Designed and wrote a pipeline to apply the model to all of Nigeria, condense predictions, and report results in a readable form.

- Accessed RGB Google Satellite Images and Sentinel-2 via Python and QGIS API.
- Manual annotation of fish ponds.

Light Echo simulation pipeline (2023-present)

- Developed a pipeline to apply the forward modeling of dust and transients to generate light echo images.
- Injected simulations into DECam and Rubin Observatory data using the LSST Science Pipelines to recreate realistic sky backgrounds.

Model interpretability (2023-present)

- Developed and trained multiple model architectures to explore latent space interpretability.
- Implemented contrastive learning models (CL/BYOL) to maximize similarity between similar objects in the latent space.
- Explored latent spaces using dimensionality reduction (UMAP, t-SNE) and information content calculations to link latent features with physical attributes.

Training: Schools and Workshops

- IAIFI Summer School; Boston; MA, USA; August 2026
- 2026 ESIL Innovation Summit Information; Boulder, CA, USA; May 2026
- Deep Learning for Science School; Berkeley; California, USA; June 2025 (competitive)
- Foundation Models for Astronomy Conference (Simons Foundation - Flatiron Institute); New York City, New York; May 2025
- LSST Data Science Fellowship Program 2022-2024
- La Serena School for Data Science 2023; NOIRLab; La Serena; Chile; August 2023 (competitive, full scholarship)
- International HPC Summer School 2023; Atlanta, Georgia, USA; July 2023 (competitive, full scholarship)
- Bayesian Deep Learning for Cosmology and Time Domain Astrophysics; Paris, France; June 2022

Leadership and Service

- October 2026, Organizing Committee, Open Tools, Open Sky: Doing LSST Science with Public Rubin Data & LINCC Frameworks, Co-organized a 3-day hackathon on Public Rubin Observatory's LSST Data for LATAM participants @ Universidad de Antioquia, Colombia
- August 2026, Organizing Committee, LSST is here! DP2 Hackathon, Co-organized a 2-day hackathon on Rubin Observatory's LSST Data Preview 2 (DP2) data and tools @ UD
- 2025-2026, President – Hispanic/Latin Graduate Student Association (HLGSA) @ UD
- 2024-2026, Membership Committee – Informatics and Statistics Science Collaboration (ISSC) @ LSST-Rubin
- 2023-2024, Treasurer - Physics and Astronomy Graduate Student Society (PAGSS) @ UD

Posters and talks

- Rubin Community Workshop 2026 – LSST; SLAC, Menlo Park, CA, USA; July 2026; Talk Title: *ML reliability model for Rubin & Rubin Difference Detectives - Zooniverse project*
- Rubin Community Workshop 2026 – LSST; SLAC, Menlo Park, CA, USA; July 2026; *Machine Learning reliability model update on the Updates from Science Pipelines session.*
- Time-Domain Astronomy with LSST in NYC, April 2026, Poster title: *ML Real-Bogus model for Rubin & Rubin Difference Detectives Zooniverse Project.*
- NASRDA-IrrEO, Abuja, Nigeria; March 2026; Talk title: *Automated detection and segmentation of smallholder fish ponds in Nigeria.*
- LSST-DA Member Reps meeting, remote; November 2025; *Update on the Rubin Difference Detective Zooniverse project.*
- Data Science Symposium; Newark, DE USA; October 2025; Poster title: *Automatic Delineation of Fish Ponds across Nigeria with Machine Learning Image Segmentation Models.*

- LSST@Europe7; Poznan, Poland; September 2025; Talk title: *Creating a Light Echoes dataset with the LSST Science Pipelines*.
- Rubin Community Workshop 2025 – LSST; Tucson, AZ, USA; July 2025; *Machine Learning reliability model update on the Updates from Science Pipelines session*.
- Rubin Community Workshop 2025 – LSST; Tucson, AZ, USA; July 2025; Talk Title: *Creating a Light Echoes dataset with the LSST Science Pipelines*.
- American Geophysical Union (AGU) Fall Meeting 2024; Washington DC, USA; December 2024; eLightning Presentation Title: *Automatic Delineation of Fish Ponds across Nigeria with Machine Learning Image Segmentation Models*.
- Rubin Community Workshop 2024 – LSST; SLAC, Menlo Park, CA, USA; July 2024; Poster Title: *A comparative study of different Deep Learning image-based models for Real-Bogus classification*.
- 2023 Annual Meeting of the APS Mid-Atlantic Section; Newark, DE, USA; November 2023; Talk title: *Forward modeling of dust and transients - a method for the generation of synthetic Light Echoes*.
- LSST@Europe5: Towards LSST Science, together!; Poreč, Croatia; September 2023; Poster title: *Forward modeling of dust and transients - a method for the generation of synthetic Light Echoes*.
- Data Science Symposium; Newark, DE USA; September 2023; Poster title: *Detangling the Mysteries of a CNN Used to Separate Astrophysical Transients from Artifacts*.
- International HPC Summer School; Atlanta, Georgia, USA; July 2023; Poster title: *The potential for Convolutional Neural Networks for transient detection without template subtraction*.
- Bayesian Deep Learning for Cosmology and Time Domain Astrophysics; Paris, France; June 2022; Poster title: *There's no difference: Convolutional Neural Networks for transient detection without template subtraction*.
- Data Science Symposium; Newark, DE USA; November 2021; Poster title: *There's no difference: CNN for transient detection without template subtraction*.

Work Experience

Junior BI Analyst, Minister of Mines and Energy

March - August 2021

Ministerio Minas y Energía, Bogotá, Colombia

Supervisor: Julian Paéz.

Languages

Spanish: Mother Language

English: Fluent